1) Describe the difference between Maximum Likelihood Estimation (MLE) and Maximum A Posterori (MAP) Estimation in one sentence.

Dude... Seriously, <u>ONE</u> sentence?

MAP estimation is systemitized Bayes Theorem; MLE is the special case when your priors don't apply.

Seriously?

2) Suppose your prior is a normal distribution centered on zero, and a variance of one,

$\theta \sim N(0,1)$

Which parameter is concidered more plausible before seeing any data, and why?

- 0.2
- 8

It represents a skeptical mental stance. Our beliefs tend toward zero without evidence to the contrary; therefor more credence should be on the null hyphothisis.

3) Suppose you only have <u>three</u> examples.

examples. Would MAP differ more or less than MLE than if you had 3 million examples?

MAP would differ less with MLE after 3 mil examples, because we align our priors to more match the data. We wouldn't think much of three crits in a row on a d20, but if it's every night, something is fishy with that die.

4) Why does a Gaussian prior produce an L2 penalty, rather than an L1?

In the space normal distributions, distances are spherical (Euclidian). Baye's Theorem says that if a credence is zero, it can never rise, but it's relatively easy to raise a low credence given good evidence.

That's why it's helpful to think of covariant data to be generally spherical, or ellipseoid when it isn't whitened. A Gaussian probability has an e^{-x^2} factor, which under log is $-x^2$.

An L1 penalty tends to have constant pressure on values close to zero, so they tend to go all the way to zero.

5) Suppose instead you chose a distrubution with a much sharper peak at zero, and heavier tails, such as a Laplase distribution. Without worrying about formulas, what kind of model behavior do you think would encourage compared to a Gaussian distribution?

When we want to weed out possibilities, the L1 penalty tends to bring probabilities all the way to zero. It also tends to accept weak evidence more strongly.